

1. Sort the dataset by attribute X :

Arrange all examples (x_j, y_j) in ascending order of X .

This ensures candidate splits are found only between distinct consecutive values of X . Sorting cost is $O(n \log n)$.

2. Initialize midpoint set:

Create an empty set M to store all possible split points. A split point is the midpoint:

$$v = \frac{x_j + x_{j+1}}{2}$$

between two successive distinct values x_j and x_{j+1} .

3. Initialize class counts:

Let n_i be the running count of examples seen so far for each class c_i , $i = 1, \dots, k$. Initially, all $n_i = 0$.

4. Scan the sorted dataset to find midpoints and counts:

For each position j from 1 to $n - 1$:

- If $y_j = c_i$, increment n_i by 1 (class count update).
- If $x_{j+1} \neq x_j$, compute the midpoint $v = (x_j + x_{j+1})/2$ and add it to M .
This ensures only distinct candidate splits are stored.

5. Record N_{vi} values:

For each class c_i , store:

$$N_{vi} = n_i$$

where N_{vi} is the number of points such that $x_j \leq v$ and $y_j = c_i$ (Eq. 19.12).

6. Handle the last example:

If $y_n = c_i$, increment n_i to ensure counts are complete.

7. Initialize best split tracker:

Set:

$$v^* \leftarrow \emptyset, \quad \text{score}^* \leftarrow 0$$

(For Gain, higher is better; for Gini/CART, start with ∞ and look for smaller scores.)

8. Evaluate each candidate midpoint $v \in M$:

- Compute class PMFs for the left partition D_Y (where $X \leq v$):

$$\hat{P}(c_i | D_Y) = \frac{N_{vi}}{\sum_{j=1}^k N_{vj}} \quad (\text{Eq. 19.14})$$

- Compute class PMFs for the right partition D_N (where $X > v$):

$$\hat{P}(c_i | D_N) = \frac{n_i - N_{vi}}{\sum_{j=1}^k (n_j - N_{vj})} \quad (\text{Eq. 19.16})$$

- Calculate split score:

Using Eq. (19.5) for Gain:

$$\text{score}(X \leq v) = \text{Gain}(D, D_Y, D_N)$$

Gain prefers splits where class distributions differ significantly between D_Y and D_N .

9. Update best split if current score is higher:

If:

$$\text{score}(X \leq v) > \text{score}^*$$

then set:

$$v^* \leftarrow v, \quad \text{score}^* \leftarrow \text{score}(X \leq v)$$

10. Return the best split and score:

Output:

$$(v^*, \text{score}^*)$$

where v^* is the best midpoint split for attribute X .